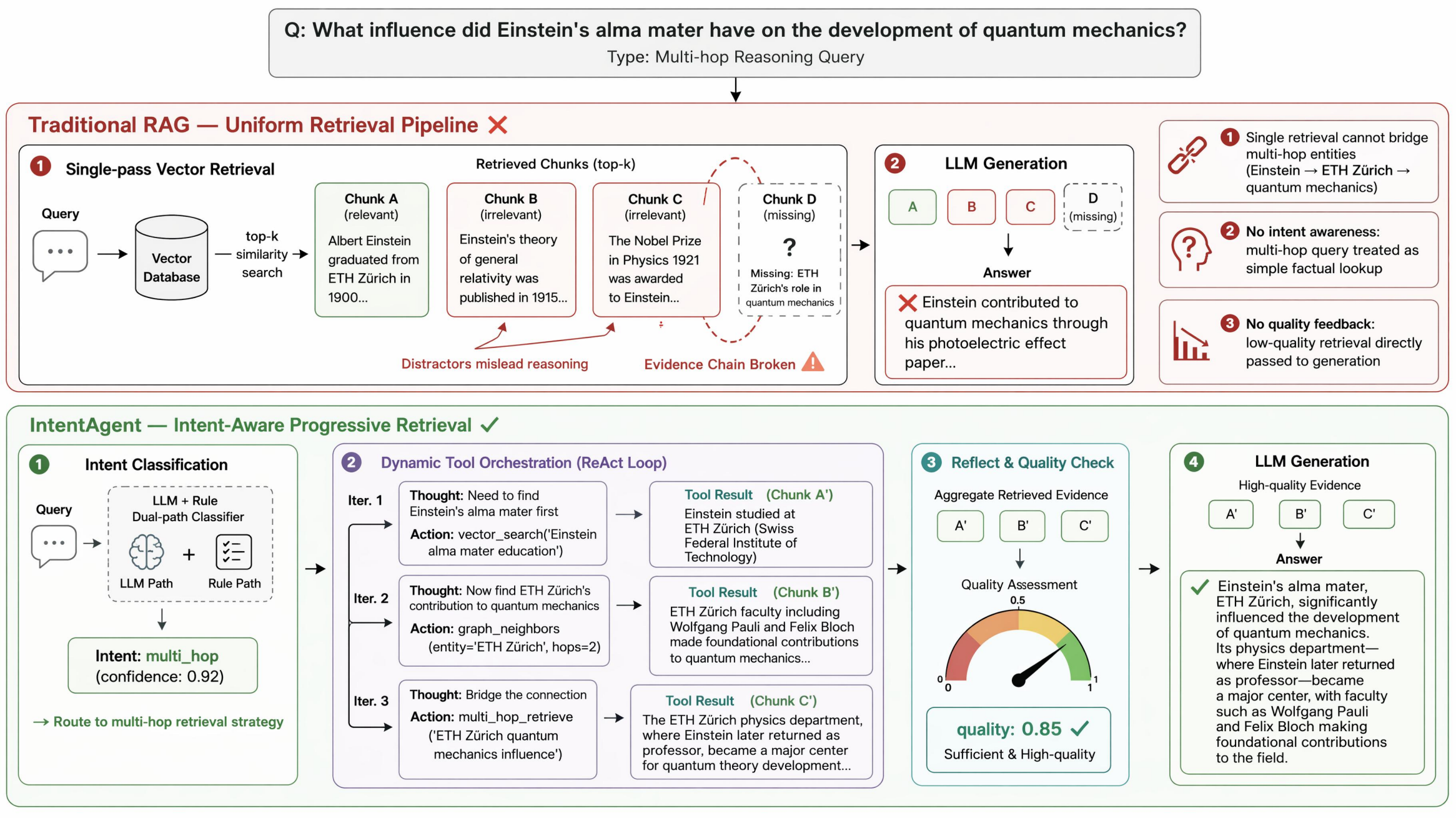

Q: What influence did Einstein's alma mater have on the development of quantum mechanics?
Type: Multi-hop Reasoning Query
Traditional RAG — Uniform Retrieval Pipeline ✗
1 Single-pass Vector Retrieval
Query
Vector Database
top-k similarity search
Retrieved Chunks (top-k)
Chunk A (relevant)
Albert Einstein graduated from ETH Zürich in 1900...
Chunk B (irrelevant)
Einstein's theory of general relativity was published in 1915...
Chunk C (irrelevant)
The Nobel Prize in Physics 1921 was awarded to Einstein...
Chunk D (missing)
?
Missing: ETH Zürich's role in quantum mechanics
Distractors mislead reasoning
Evidence Chain Broken
2 LLM Generation
A
B
C
D (missing)
Answer
✗ Einstein contributed to quantum mechanics through his photoelectric effect paper...
1 Single retrieval cannot bridge multi-hop entities (Einstein → ETH Zürich → quantum mechanics)
2 No intent awareness: multi-hop query treated as simple factual lookup
3 No quality feedback: low-quality retrieval directly passed to generation
IntentAgent — Intent-Aware Progressive Retrieval ✓
1 Intent Classification
Query
LLM + Rule Dual-path Classifier
LLM Path
+
Rule Path
Intent: multi_hop (confidence: 0.92)
→ Route to multi-hop retrieval strategy
2 Dynamic Tool Orchestration (ReAct Loop)
Iter. 1
Thought: Need to find Einstein's alma mater first
Action: vector_search('Einstein alma mater education')
Tool Result (Chunk A')
Einstein studied at ETH Zürich (Swiss Federal Institute of Technology)
Iter. 2
Thought: Now find ETH Zürich's contribution to quantum mechanics
Action: graph_neighbors (entity='ETH Zürich', hops=2)
Tool Result (Chunk B')
ETH Zürich faculty including Wolfgang Pauli and Felix Bloch made foundational contributions to quantum mechanics...
Iter. 3
Thought: Bridge the connection
Action: multi_hop_retrieve ('ETH Zürich quantum mechanics influence')
Tool Result (Chunk C')
The ETH Zürich physics department, where Einstein later returned as professor, became a major center for quantum theory development...
3 Reflect & Quality Check
Aggregate Retrieved Evidence
A'
B'
C'
Quality Assessment
0
0.5
1
quality: 0.85 ✓
Sufficient & High-quality
4 LLM Generation
High-quality Evidence
A'
B'
C'
Answer
✓ Einstein's alma mater, ETH Zürich, significantly influenced the development of quantum mechanics. Its physics department—where Einstein later returned as professor—became a major center, with faculty such as Wolfgang Pauli and Felix Bloch making foundational contributions to the field.